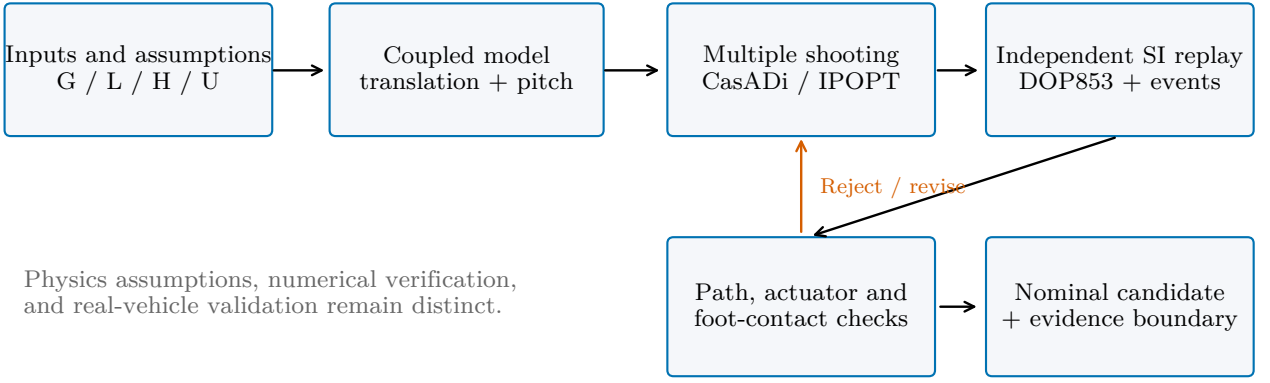
Inputs and assumptions
G / L / H / U
Coupled model
translation + pitch
Multiple shooting
CasADi / IPOPT
Independent SI replay
DOP853 + events
Reject / revise
Path, actuator and
foot-contact checks
Nominal candidate
+ evidence boundary
Physics assumptions, numerical verification,
and real-vehicle validation remain distinct.